



# SOLAR-POWERED IoT BASED ROBOT FOR RAILWAY TRACK INVESTIGATION AI ASSISTANCE

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## INTRODUCTION

- ✓ This system uses IoT and sensor-based technologies to detect obstacles on railway tracks in advance.
- ✓ It allows real-time monitoring through cameras and GPS, sending live data to the train control system for quick decision-making.
- ✓ The robot operates using solar power and can be remotely controlled via mobile or laptop, ensuring efficient, eco-friendly, and reliable railway safety.

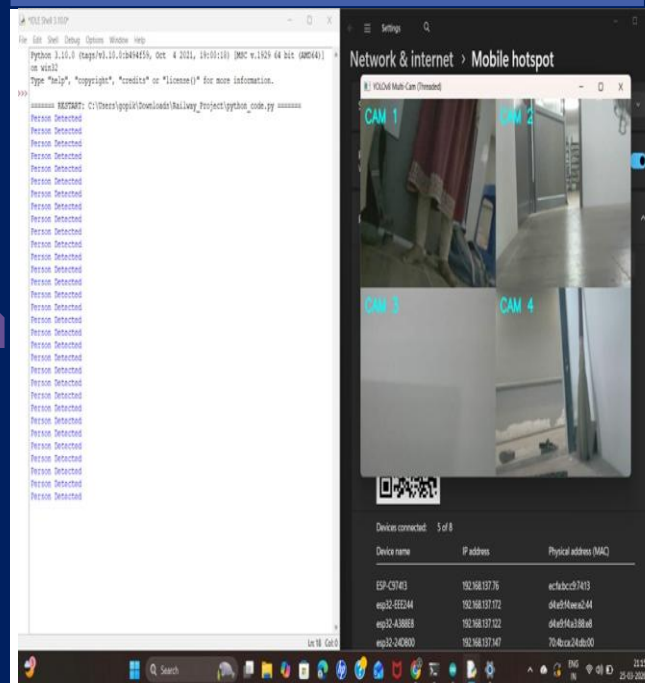
## ADVANTAGES

- ✓ Proactive safety with early obstacle detection.
- ✓ 360° monitoring using cameras
- ✓ AI-based obstacle classification
- ✓ Solar-powered and energy efficient

## COMPONENTS

- ✓ ESP8266 NodeMCU
- ✓ IR Sensor
- ✓ Ultrasonic Sensor
- ✓ ESP32-CAM Module
- ✓ GPS Module
- ✓ LCD Display (16x2)
- ✓ Buzzer
- ✓ Solar Panel

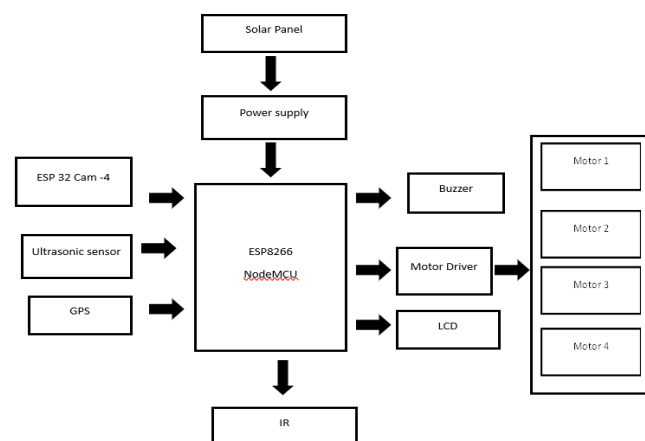
## OUTPUT



## FUTURE SCOPE

- ✓ In the future, the system can be enhanced with advanced AI for automatic obstacle detection and faster decision-making.
- ✓ It can be upgraded to work autonomously without manual control.
- ✓ High-resolution and thermal cameras can improve performance in night and bad weather conditions.
- ✓ The system can also be linked with train braking mechanisms for instant response.

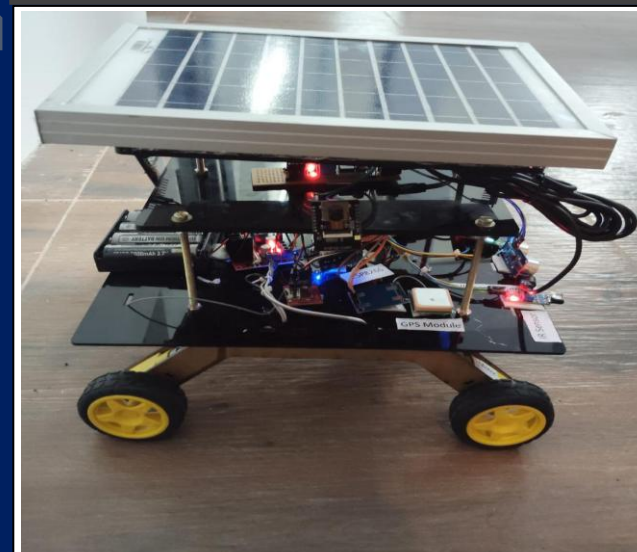
## BOLCK DIAGRAM



## APPLICATIONS

- ✓ Railway track safety for passenger and freight trains.
- ✓ Early detection of animals or humans on railway tracks.
- ✓ Integration with railway traffic management systems for automated alerts
- ✓ Could be expanded for industrial track monitoring or mining railways.

## PROTOTYPE



## CONCLUSION

- ✓ This project provides a smart and efficient solution to improve railway safety using a solar-powered IoT-based robot.
- ✓ It detects obstacles early and offers real-time monitoring through cameras and GPS.
- ✓ The system reduces accidents by giving timely warnings to train operators.
- ✓ Its eco-friendly design ensures reliable and continuous operation.